

TSW6223 Semantic Web Technology

Title: Semantic Web Technology for Solving Real-World Problem

(Feel free to change the title to fit your work)

Introduction:

Semantic web technology (SWT) enhances internet data by making it more structured, machine-readable, and interconnected, thereby supporting improved automation, search, and information integration. Its core objective is to enable computers to interpret the meaning of data rather than merely presenting it as static content. This group project aims to examine the practical applications of SWT across a range of domains, including e-commerce, education, banking and finance, entertainment, agriculture, management, healthcare, data science, law, and engineering. For example, SWT facilitates seamless data sharing across different platforms and domains. In business contexts, it supports the automation of complex processes such as maintaining links and product databases, integrating customer data across multiple systems, and executing transactions based on well-defined rules. Additionally, SWT plays a key role in managing and interpreting organisational data by promoting standardised formats and vocabularies, ensuring consistency, traceability, and clarity.

Objectives:

1. To understand the core concepts and underlying principles of SWT.
2. To design and implement practical SWT-based solutions for two selected topics from Categories 1 to 3.

Project Phases:

1. Topic of Selection and Proposal

This is a group project. The number of group members is **at most 4 students**. Each group comprises members with diverse skills in programming, data management, and domain knowledge relevant to the chosen topics. All members in a group shall register their names and obtain a group ID (link: [T2610 TSW6223 Semantic Web Technology](#)).

Each group will choose **two** semantic web technologies and tools from the following categories:

- (a) Category 1: Extensible Markup Language (XML)
- (b) Category 2: Resource Description Framework (RDF), RDF with Schema (RDFS), and SPARQL Protocol and RDF Query Language
- (c) Category 3: Web Ontology Language (OWL) and/or inference

2. Research and Analysis

- (a) Students will conduct research in details to understand the core concepts and functionalities of the selected topics.
- (b) Students explore real-world applications where the chosen topics can be effectively applied to solve practical problem(s).
- (c) **Project leader shall submit a project title (by 4pm Thursday Week 3 through [T2610 TSW6223 Semantic Web Technology](#)).**

3. Application Development

- (a) Students shall develop an application that leverages the selected SWT and tools.

- (b) Students implement the designed solutions using appropriate programming languages (e.g., Python, Java, C#, C/C++).

4. Evaluation

Conduct functionality testing on the developed solutions to ensure they meet the specified requirements and perform as expected.

5. Documentation

Each group will prepare **one** comprehensive report describing the project and problem statement, explanation of the SWT used (XML, RDF, RDFs, SPARQL and/or OWL), solution design, implementation details, and evaluation results. Students shall also explain future improvements and potential extensions of the application.

Each group member shall indicate the portion of writeup in the report. All facts shall be cited with reference(s) in the report. References are cited in the APA style.

6. Presentation and Demonstration

All students will deliver a presentation summarising their project, highlighting key insights, demonstrate the functionality and technical details of the proposed solutions, and attend a Q&A session. Group leader shall book a presentation time slot on group members (the details will be informed in Week 10).

Assessment Criteria:

Assessment criteria are based on understanding of SWT concepts, creativity and innovation in solution design, effectiveness of the solutions in addressing real-world problems, technical complexity, quality of report and presentation, demonstration and teamwork within the group. Please refer to the rubric for details.

Submission:

1. Materials to Submit

A **file folder** (naming format [GroupXX, for an example, Group05](#)) in the Project Leader's Microsoft OneDrive that contains (i) a softcopy of report in the docx and pdf format* (ii) source codes/solution (iii) presentation slides (iv) a plagiarism report assessing the contents in (i) that is downloaded from the Turnitin website.

* after being checked with Turnitin, assure that overall similarity score is at most 20%.

2. Submission

Group leader shall complete an online submission form by **4pm Friday Week 13 (26 June 2025)**. Share the link with Prof. Dr. Tan Shing Chiang (sctan@mmu.edu.my)

3. Group Presentation and Demonstration

It is scheduled in the lab sessions in **Week 14 (29 June – 5 July 2025)**. Each member must participate in presentation and make a demonstration.

Report Format:

1. First page

TSW6223 SEMANTIC WEB TECHNOLOGY
PROJECT REPORT
TERM 2610

Title:

Group ID: *(Group leader will register on behalf of all members.)*

Selected Topics (1) Category 2 XXX (2) Category 3 XXX

Members:

Name/ID	List contribution(s) in the project	State which part of the write-up in the report
1 (Project Leader) Patrick Lim Wei Jie /21110000		
2		
3		
4		

2. Second page and onwards

1. Introduction
2. Problem Statement and Objectives
3. Solution Development
4. Evaluation
5. Future Improvements
6. Conclusion
7. References (APA style)

Rubric (40 marks)**A. Report Quality (20 marks)****1. Knowledge and Understanding (5 marks)**

Score	Note
5	The report demonstrates a deep understanding of the Semantic Web technologies used (XML, RDF, OWL, SPARQL, etc.). The student not only describes these technologies but also demonstrates how and why they were chosen for the project, showing critical thinking and the ability to apply the technologies effectively.
4	The report shows a solid understanding of the core technologies and their application. The student explains their choices and demonstrates the use of the technologies, although some aspects could benefit from more in-depth explanation or examples.
3	The report shows a general understanding of the technologies, but there are some gaps or misunderstandings. The explanation is somewhat superficial, and some critical details about the technologies' application may be lacking.
2	The report demonstrates limited understanding of the core technologies. The explanation is basic, with little connection between theory and application, or some technologies are not explained properly.
1	The report reflects minimal understanding of the Semantic Web technologies. There is confusion in explaining their application, and critical aspects are missing or incorrect.
0	The report shows little to no understanding of the core technologies. The concepts are misunderstood or misapplied, and the explanation lacks coherence.

2. Organisation (5 marks)

Score	Note
5	The report is well-organized and logically structured. Each section flows smoothly into the next, and the content is grouped logically. The report includes all necessary sections (e.g., introduction, methodology, results, conclusions)
4	The report is mostly well-organized. Sections are logically structured, but there may be minor issues with the order of content or some sections could be more clearly defined.
3	The report is somewhat organized, but there are issues with the flow or structure. Some sections might seem out of place or lack sufficient detail.
2	The report lacks proper organization. Sections are poorly structured or missing key parts, and the flow of ideas is difficult to follow.
1	The report is disorganized, with little to no logical structure. Sections are scattered, and it is hard to discern the main points.
0	The report is completely disorganized and lacks any clear structure or sections.

3. Format (5 marks)

Score	Note
5	The report adheres to the required format. It includes a clear title, table of contents, proper headings and subheadings, and uses consistent font size, style, and spacing. References and citations are formatted correctly.
4	The report generally follows the required format. Some minor formatting issues (e.g., inconsistent headings, minor citation errors) may be present.
3	The report mostly follows the required format, but there are noticeable formatting errors. Headings, spacing, or citations may be inconsistent or incorrect.
2	The report does not fully follow the required format. Formatting issues such as missing headings, incorrect citations, or inconsistent font styles are present.
1	The report fails to follow the required format. There are significant formatting issues, making the report look unprofessional.
0	The report does not adhere to any standard format or guidelines, making it difficult to follow or unreadable.

4. Innovation (5 marks)

Score	Note
5	The project is highly innovative, presenting a unique or novel use of Semantic Web technologies. The solution or application goes beyond typical implementations and demonstrates creativity in solving real-world problems.
4	The project demonstrates creativity and original thinking. While it may not be entirely novel, the use of Semantic Web technologies in the application is somewhat innovative, with a solid and practical solution.
3	The project is somewhat innovative, but the application of Semantic Web technologies may be standard or lack originality. There are some creative elements, but the overall approach is not groundbreaking.
2	The project is not particularly innovative and follows basic or common solutions to a problem. The use of Semantic Web technologies is typical and lacks unique or creative application.
1	The project lacks innovation, using standard solutions with little to no new ideas or creative application of Semantic Web technologies.
0	No innovation demonstrated; the project is derivative, copying existing solutions without any unique application or contribution.

B. Presentation (5 marks)

Score	Note
5	The presentation is well-structured, engaging, and clear. It effectively communicates the problem, the solution, and the key technologies used. Visual aids (slides, demos) are clear and support the narrative. The group handles questions confidently and provides thorough answers.
4	The presentation is clear and structured, covering all major aspects. Some areas may be less engaging, but the message is conveyed well. Visual aids are relevant but may need improvement. The group handles questions well, though some responses may be less thorough.
3	The presentation is somewhat disorganized or unclear. Some important details are omitted, and the visuals may be hard to follow. The group handles some questions well, but some answers may lack depth or clarity.
2	The presentation lacks clear structure or is difficult to follow. Important aspects of the project are left out. Visual aids are minimal or poorly designed, and the group struggles to answer questions or provides incomplete answers.
1	The presentation is poorly organized, hard to follow, or lacks key details. Visual aids are irrelevant or absent. The group struggles to communicate the project's purpose and has difficulty answering questions.
0	No presentation given or the presentation fails to meet the basic requirements.

C. Clarity in Software (5 marks)

Score	Note
5	The software is clear, well-documented, and easy to understand. The integration of XML, RDF, OWL, and/or SPARQL is seamless. Code is clean, efficient, and adheres to best practices. Well-commented with clear explanations of key sections.
4	The software is mostly clear and functional, with some areas that may require additional documentation or explanation. The integration of technologies is solid, but there may be minor inconsistencies or inefficiencies in the code.
3	The software is somewhat unclear or lacks adequate documentation. Some integrations may be confusing or inefficient, and there may be sections of the code that are not well-explained or commented.
2	The software is difficult to understand, with significant issues in clarity or documentation. Integration of the technologies is incomplete or confusing. Code may not work as intended or is poorly organized.
1	The software is poorly written, lacks clarity, and is difficult to navigate. There is little to no documentation, and key functionalities are missing or broken.
0	No software submitted or the software is not functional.

D. Demonstration (5 marks)

Score	Note
5	The demonstration is clear, well-organized, and showcases the key functionalities of the application effectively. All aspects of the project are demonstrated, and the software works as intended. The group is able to handle questions and troubleshoot on the spot.
4	The demonstration is clear, but some aspects may not be fully showcased. The software works well, with minor issues that are quickly resolved. The group answers questions adequately.
3	The demonstration is somewhat unclear or incomplete, with a few features not shown. Some parts of the software may not function as expected, and the group struggles to answer some questions.
2	The demonstration lacks clarity or key features are missing. The software may have significant bugs or functionality issues. The group has difficulty answering most questions.
1	The demonstration is disorganized, incomplete, or the software fails to work as expected. The group is unable to effectively showcase the project or answer questions.
0	No demonstration given or the demonstration does not meet the assignment requirements.

E. Teamwork (5 marks)

Score	Note
5	The team collaborates effectively, with clear distribution of tasks and responsibilities. All members contribute meaningfully to the project, and the group works well together to solve problems and complete tasks on time.
4	The team collaborates well, with most members contributing meaningfully. Some tasks may not be equally distributed, but overall cooperation and progress are strong.
3	The team works together, but there are some issues with task distribution or coordination. Some members may contribute less, and some tasks may be incomplete or delayed.
2	Team collaboration is poor, with some members contributing very little or tasks being delayed. There may be significant communication or coordination problems within the group.
1	The team struggles to work together, with minimal collaboration. Many tasks are not completed or are poorly done, and communication is ineffective.
0	No teamwork demonstrated or the group fails to complete the project.